Developing a Machine Learning Model to Predict Fresh Frozen Plasma Requirements during Stanford Type A Acute Aortic Dissection Surgery Using Early ROTEM FIBTEM Data

Calibrated Synthetic Dataset (Phase 2)

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Motivation

Stanford type A acute aortic dissection (TAAAD) requires emergency surgery and is often complicated by severe coagulopathy leading to massive bleeding. Current European and U.S. hemostatic guidelines recommend fibrinogen concentrate or cryoprecipitate for the replacement of depleted fibrinogen[1,2]. However, in Japan, fibrinogen concentrate is not approved for this indication, and cryoprecipitate is rarely available; therefore, fibrinogen replacement mainly relies on large-volume transfusion of fresh frozen plasma (FFP), which can cause hemodilution and hypervolemia[3]. To mitigate these effects, FFP is administered during cardiopulmonary bypass (CPB) while excess fluid is removed using extracorporeal ultrafiltration (ECUM). Because FFP must be transported from the Japanese Red Cross and thawed before use, early estimation of the required volume is essential to ensure that the hemostatic requirement can be fulfilled before CPB weaning.

Contribution

We developed a machine-learning model to predict intraoperative FFP requirements during TAAAD surgery using ROTEM FIBTEM A5 values obtained immediately after CPB initiation ("A5 start"). This ROTEM-guided approach aims to support rapid, individualized transfusion planning and may help optimize both patient safety and resource utilization.

Results (Phase2, provisional)

Model trained on 1000 synthetic + 336 clinical cases (W = 12). Evaluated using 3-fold cross-validation (seed = 202).

Metric	Result	Threshold
SevereUnder_all (%)	0.15%	≤10%
SevereOver_all (%)	0.07%	≤20%
MAE_all (U)	0.84U	≤8 U

Discussion

Model Development

We developed a FIBTEM A5-based model for predicting FFP requirements by generating 1000 synthetic cases and calibrating them to clinical data, followed by the integration of clinical cases using a weighting factor (W=12). Calibration addressed components not quantifiable from prior literature—aligning A5 start to clinical distributions, applying an ECUM-related fibrinogen loss correction (k_loss = 1.08), and adding a +4U adjustment for completely thrombosed false lumen (FL=1)—thereby improving consistency with clinical judgment.

Predictive Performance

In Phase 2, 3-fold cross-validation showed very low SevereUnder_all (0.15%) and a low MAE_all (0.84U), with almost no severe over-transfusion. These results indicate that the model reliably estimates FFP requirements based on A5 start, body weight, predicted CPB time, and aortic morphology.

Limitations

Phase 3 validation will be performed once additional cases become available for retrospective collection.

Conclusion

Phase 2 showed excellent safety and accuracy.

Phase 3 will provide full external validation before clinical use.

Methods

➢ Three-Phase Framework

Phase1	Phase2	Phase3
SYNTHETIC DATA n=1,000 synthetic cases 6 rule-based features Random Forest (3-fold CV)	**ITERATIVE CALIBRATION** Using clinical cases Prediction-only refinement Rule update w/o retraining	**FINAL VALIDATION** External evaluation Final report Independent test

Key Innovation: Clinical data iteratively refines synthetic rules **without model retraining**, enabling data-leakage-free calibration

➢ Synthetic Data Generation (n=1000, seed=42)

FEATURES (6 items)

- **Weight (kg)[4]:** 65 ± 15 kg (range 35–120 kg).
- **Predicted CPB time (min)**: based on institutional times; age and entry site[5] used only to select the surgical procedure.
  - Ascending or Hemi-arch replacement: 130 min
  - Partial or Total Arch replacement: 190 min
- **A5 start (mm):** estimated from baseline fibrinogen[4], D-dimer[5], and an assumed 1500 mL dilutional volume.
- **Dissection extent[5]:** 1 = Ascending Aorta (AA) only, 2 = AA → Descending thoracic, 3 = AA → Abdominal aorta, 4 = AA→ Iliac arteries
- **False lumen status[5]:** 1=Thrombosed, 2=Partially thrombosed, 3=Patent
- **True lumen size** *(extent types 3 & 4 only)*: 1=< ½ , 2=≥ ½ of the aortic diameter

OUTCOME

- **FFP: Required FFP dose to achieve the post-protamine A5 target (post A5) =9mm[6].**

Formula[7-9]: FFP_raw = [c_eff·(9 − A5_start)_+·Weight + 0.14·Predicted CPB time·PV_end] / 200 × k_loss

FFP = 4 · ceil(FFP_raw / 4).

c_eff: Baseline A5 responsiveness (6.25 mg/kg/mm) corrected for CPB-end plasma dilution

PV_end: plasma volume at CPB termination. PV_end = ((70Weight + 800) × (1 – Ht_cpb)) / 100, Ht_cpb = 0.30

k_loss: *Loss-correction factor accounting for fibrinogen removal through Extra-Corporeal Ultrafiltration and Membrane (ECUM). k_loss was set to 1.3 to compensate for ECUM-related fibrinogen loss (in Phase1)*

➢ Evaluation Metrics & Pass Criteria

Category	Metric	Definition / Threshold	Pass Criteria
Primary	SevereUnder_all (%)	(True − Predicted) ≥ 12 U	≤ 10%
Secondary	SevereOver_all (%)	(Predicted − True) ≥ 12 U	≤ 20%
	MAE_all (U)	Mean absolute error (4-U step)	≤ 8 U

➢ Retrospective Clinical Dataset (IRB No.1087)

Inclusion Criteria

- Hyperacute Stanford Type A AAD treated with emergency surgery (pre-rebound; if onset unclear, preoperative Fbg < 400 mg/dL used to confirm hyperacute status).
- Post-protamine A5 = 7–11 mm *or* post-protamine Fbg = 160–270 mg/dL.

Exclusion Criteria

- Missing predictor variables.

Outcome Assignment

- **Pattern 1 (post-A5):** 7 mm → +4 U, 8–10 mm → 0 U, 11 mm → –4 U
- **Pattern 2 (post-Fbg):** Fbg-based mapping aligned to the same target A5 = 9 ± 2 mm, applying the same ±4 U / 0 U rules.

Phase Allocation (seed = 202)

- **Phase 2:** Pattern 1 (25 cases) + all Pattern 2 (3 cases)
- **Phase 3:** Pattern 1 (17 cases; additional cases will be included in future evaluation)

➢ Model Development

Model: Random Forest Regressor (scikit-learn), seed = 202

Hyperparameters (Phase 1, fixed): n_estimators = 400, max_depth = None, min_samples_leaf = 1

Calibration (Phase 2): A5 start (clinical distribution), k_loss = 1.08, Morphology +4 U (FL = 1)

Training: 1000 calibrated synthetic + 336 clinical (28 cases × W=12)

References

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